

Project 2 Report

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Course: COSC 4397 - Generative AI

Part 1: Supervised Fine-Tuning (SFT) using LoRA

Prompt Template

Sort the following objects from BIGGEST to SMALLEST.

Objects: {sample}

Please provide your reasoning step by step, then output the sorted list.

OUTPUT:

Metrics

Metric Score

FCR	0.7500 (15/20)
OPA	0.0100 (1/100)
EMR	0.0000 (0/20)

Analysis

This is due to the fact that the performance of the SFT-trained model was 75% format compliance, which shows that the model was able to learn the reasoning format from the cot column. Object order performance was 1%, showing that the model had difficulty ordering objects based on their sizes. There was an exact match of 0%, which means that no sample was able to achieve format and sorting order at the same time.

Part 2: Direct Preference Optimization (DPO) using LoRA

Prompt Template

Sort the following objects from BIGGEST to SMALLEST.

Objects: {sample}

OUTPUT:

DPO Triplet Format

Field	Source
Prompt	System instructions + sample
Chosen	cot column
Rejected	reject column

Metrics

Metric	Score
FCR	0.9500 (19/20)
OPA	0.2100 (21/100)
EMR	0.0000 (0/20)

Analysis

The DPO-trained model increased the format success rate to 95% and OPA to 21% outperforming the SFT model, but still maintaining the EMR at 0%. It can be seen that the DPO method was effective in making the model like good outputs better than the bad ones, thus increasing its skills in formatting and reasoning more than the SFT-only method.

Part 3: SFT-DPO Sequential Fine-Tuning with Complex Constraints

Prompt Template

Sort these objects from BIGGEST to SMALLEST.

Objects: {sample}

Example of the correct output format:

OUTPUT:

1. [ABSOLUTE.LARGEST]: largest object
2. [SECOND.LARGEST]: second largest object
3. [MEDIAN.VOLUME]: middle object
4. [SECOND.SMALLEST]: second smallest object
5. [ABSOLUTE.SMALLEST]: smallest object

Now sort these objects:

OUTPUT:

1. [ABSOLUTE.LARGEST]:
2. [SECOND.LARGEST]:
3. [MEDIAN.VOLUME]:
4. [SECOND.SMALLEST]:
5. [ABSOLUTE.SMALLEST]:

Metrics

Metric Score

FCR	1.0000 (20/20)
OPA	1.0000 (100/100)
EMR	1.0000 (20/20)

Analysis

The sequential SFT-DPO model scored perfectly on all three metrics, with 100% FCR, 100% OPA, and 100% ECR. That marks a great improvement compared to the performance of the individual models where SFT only got 75% FCR and 1% OPA and DPO 95% FCR and 21% OPA. The sequential model managed to maintain the spatial reasoning ability of the model while adapting to the new complex format tagging requirements of Part 3. The performance of the SFT-DPO pipeline was greater than that of the DPO model alone, proving once again that sequential fine-tuning works excellently at embedding both comprehension of content and format adherence into the model. The findings show that the combination of SFT and DPO makes a powerful alignment pipeline for teaching a small language model not just what to reason about but also how to format its answers.

Final Summary Table

Part	Strategy	FCR	OPA	EMR
1	SFT	0.7500	0.0100	0.0000
2	DPO	0.9500	0.2100	0.0000
3	SFT-DPO	1.0000	1.0000	1.0000

Conclusion

SFT-DPO alignment made it possible to turn the model from a generic statistical predictor of words into a reasoning machine that is able to sort objects based on physical size according to structured reasoning and high format compliance.

At the first step, SFT training gave us baseline competence in terms of learning how to reason and use the pre-generated cot format, which scored 75% of format compliance. Nevertheless, the low accuracy of positioning of objects in space shows that though the format was learned, the real spatial reasoning skills were not acquired. The second step of training, DPO, improved not only the format compliance (95% FCR) but also object position accuracy in 21% OPA. Thus, it can be concluded that the model learned how to reinforce correct patterns of reasoning and punish incorrect reasoning and format problems.